XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

Department of Electrical and Electronic Engineering

Xiamen University Malaysia

Quiz 1 (Q4)

EEE407 Electrical Energy Utilization

INSTRUCTIONS TO CANDIDATES

1. Use Microsoft Excel to solve this question, and upload the file to Moodle.
2. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you will be penalized accordingly.

Question 4 [CLO3-PLO2:C5] (30 marks)

1. To reduce energy usage, the management of an office building launched an energy project. New LED lamps. air-conditioning systems, and other electrical systems were upgraded in phases, specifically during the first two months of 2020. The building's energy consumption from January 2019 to February 2021 was recorded, as shown in Table 2. Additionally, monthly independent variables such as cooling degree days and the number of staff, which could influence the building's energy consumption, were included.
   a. Choose the appropriate baseyear, then use regression method to determine the correlations between energy consumption and the two independent variables: CDD and number of staff. Plot the linear model and show the baseline equation and R^2 value for both independent variables. (10 marks)
   b. Choose the suitable linear model, then determine the adjusted energy consumption (baseline energy) for each month and the monthly cumulative energy savings after the installation period. Show all data in a table and plot the energy profile (kWh vs. month) consists of all the stages: pre-retrofit, installation, post-retrofit, and baseline energy. (8 marks)

c. Estimate the simple payback period in years if the average energy cost is $0.50/kWh, and the total cost of the energy project is $500,000. (2 marks)

Table 2: Energy consumption from January 2019 to February 2021.

Month-year	Cooling degree days (CDD)	Number of staff	Energy consumption (kWh)
Jan-19	970	510	34995
Feb-19	957	508	33765
Mar-19	945	513	32466
Apr-19	938	510	32573
May-19	910	511	30917
Jun-19	882	505	27164
Jul-19	875	505	26257
Aug-19	853	503	28112
Sep-19	893	498	30625
Oct-19	837	501	24690
Nov-19	863	496	29781
Dec-19	920	495	31127
Jan-20	959	56	10437
Feb-20	936	61	9786
Mar-20	840	490	12381
Apr-20	816	491	10214
May-20	894	505	19578
Jun-20	901	500	19160
Jul-20	848	499	15352
Aug-20	926	511	20485
Sep-20	998	512	25242
Oct-20	880	512	18708
Nov-20	884	510	21816
Dec-20	882	490	16610
Jan-21	869	495	11935
Feb-21	946	510	21565